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Supervisor: Dr. Ing Georg Hasemann

Beyond Nickel: The Physics of BCC-Superalloys

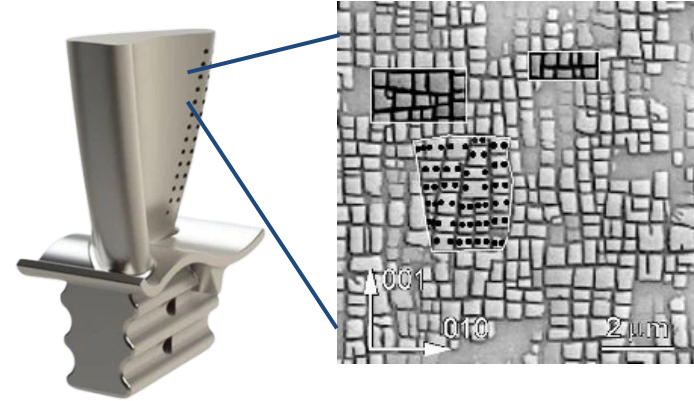
Microstructural Mechanics & Deformation

Team Project Presentation | Materials Science MSI

Ni-based Superalloys and the need of BCC structures

FCC Ni-Based Superalloys

- Approaching fundamental thermodynamic limit
 - Ni melting temperature **1455 °C**
 - Low solvus temperature $\approx$ **1300 °C**

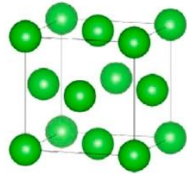
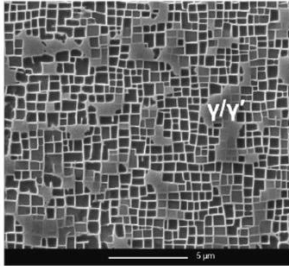


BCC Refractory Superalloys

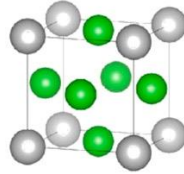
- The ultra-high temperature frontier (>1300 °C to 1800 °C)
- Utilizing **refractory elements** (W, Mo, Nb, Ta) paired with ordered precipitate formers (Al, Ti, Zr)

Refractory Mechanisms in Ni-based Superalloy to BCC-Superalloy

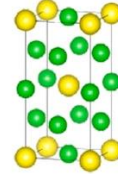
FCC Ni-base
superalloys



γ A1 Ni solid solution

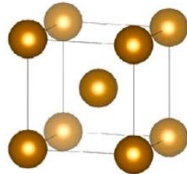
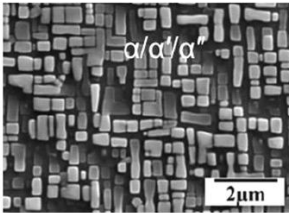


γ' L1₂ ordered Ni₃Al

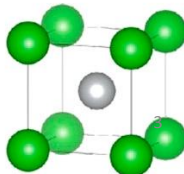


γ'' D0₂₂ ordered Ni₃Nb

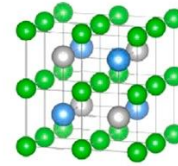
BCC Fe-base
superalloys



α A2 Fe solid solution



α' B2 ordered NiAl



α'' L2₁ ordered Ni₂AlTi

The core concept: A disordered BCC (A1) matrix reinforced by coherent, ordered superlattice precipitates (L1₂)

The analogy: Replacing the traditional γ/γ' architecture with a high temperature A2 matrix with B2/L2₁ precipitates 3

Project scope & Workflow

A comprehensive literature review is required to synthesize and articulate the refractory mechanisms inherent to BCC-Superalloys, as well as engineering factors.

Step 1: Literature Synthesis - aggregating recent Scripta Materialia finding on the nascent BCC-superalloy class.

Step 2: Review deformation mechanisms of non-refractory alloys and Ni-based superalloy

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Step 3: Discuss the deformation mechanics of Fe-based superalloys and BCC-RHEA

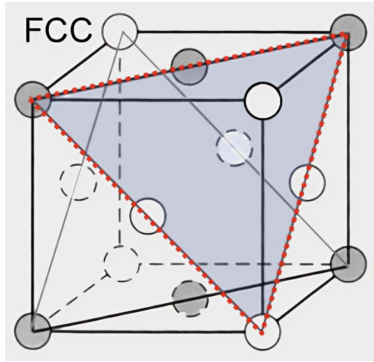
Step 4: Engineering outlook (Oxidation, Additive manufacturing, etc.)

Outlines

- Ni-based superalloys
- Project & Workflow
- **Fundamental Physics & Fe-based BCC Superalloys**
 - Dislocation Core Topology
 - Kink-Pair Mechanism
 - Fe-Based BCC Lattice Misfit
 - Low Lattice Misfit
 - Hierarchical Microstructure (Moderate to High Lattice Misfit)
- Dislocation-Precipitate Interactions: Shearing vs Orowan Looping
- Refractory High Entropy Superalloys
 - Core High Entropy effects
 - Modified deformation characteristic: Matrix ruggedness
 - Pathway engineering
 - Engineering challenges & Outlooks

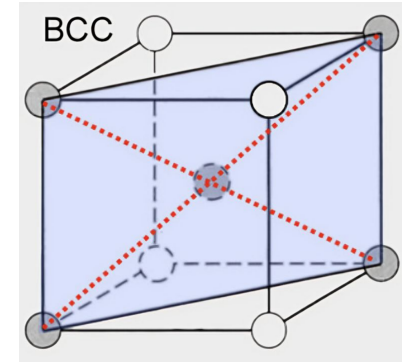
FCC vs BCC: Dislocation Core Topology

FCC Lattices (APF=0.74)



- High atomic packing, slow diffusion
- Deformation on close packed $\{111\}$ planes
- Perfect dislocation dissociates into Shockley partials
- 2D planar core $\rightarrow$ low Peierls stress
- Dislocation pinned to a single plane.
- Yield strength temperature-independent

BCC Lattices (APF=0.68)

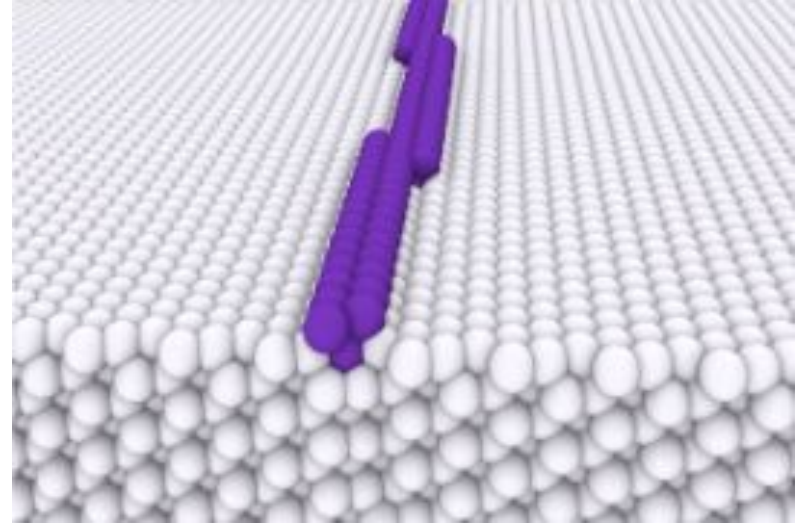


- More open, no true close-packed planes
- Screw dislocation ($b=a/2 \langle 111 \rangle$).
- Non-planar 3D core spreading on $\{110\}$
- High intrinsic lattice resistance
- Thermally activated kink-pair mechanism

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BCC Kink-Pair Mechanics

- Non-planar core is locked → Require thermal activation for motion
- The kink-pair mechanism:
 - Thermal activation: Thermal fluctuations provide energy to small segments of the screw dislocation to overcome the Peierls barrier, nucleating a pair of kinks.
 - Kink propagation: Edge-character kinks are highly mobile → spread laterally
- Result:
 - Temperature dependence
 - Screw-controlled plasticity
 - High ductile-to-brittle transition temperature → limits room-temperature applications



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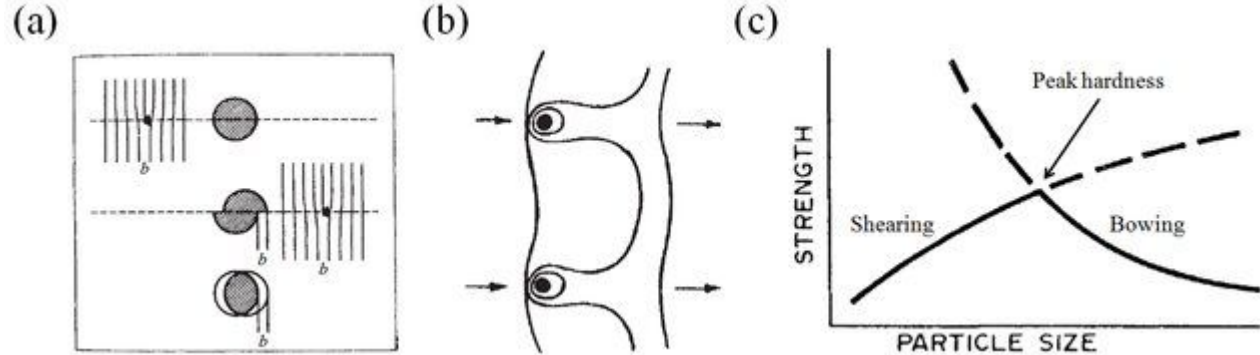
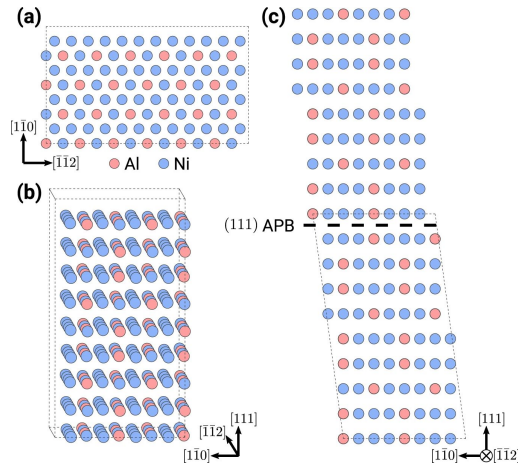
Dislocation-Precipitate Interactions: Shearing vs Orowan Looping

- Particle Shearing:

- Dislocation cuts directly through precipitates
- Create Anti-Phase Boundary (APB) defect
- Dislocation pair up (super-dislocation) to erase APB
- FCC: up to 20 nm
- Ferritic superalloy: $< 3 \text{ nm} \rightarrow$ high APB energy
- RHEA: extend to 50 nm

- Orowan Looping:

- Dislocation bows around particle, leaves Orowan loop
- Dominant in BCC for typical sizes (30 - 54 nm)
- Coarsening $\rightarrow$ strength lost



Outlines

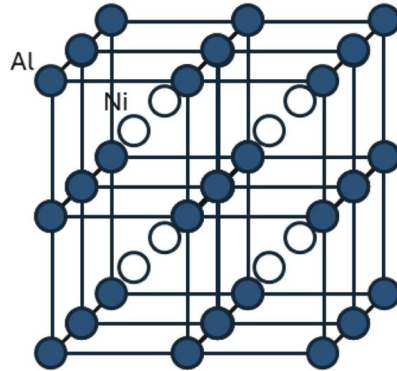
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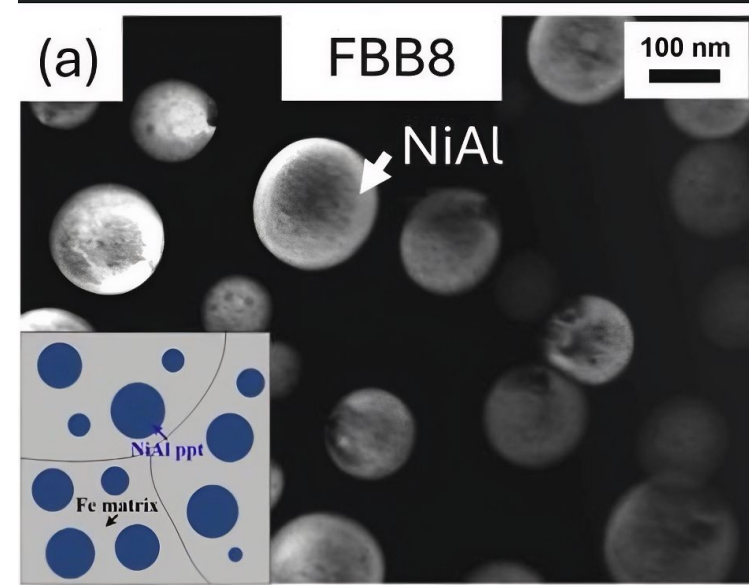
Fe-based BCC Lattice Misfit - FBB8 (Fe-Ni-Al-Cr)

Low Misfit (<0.5%)

- Create spherical, coherent B2 precipitates.
- Minimal thermodynamic driving force for coarsening ensure long-term thermal stability
- Lack of elastic coherency strain renders them mechanically weak against dislocation climb at high temperature



Eight unit cells of NiAl-B2

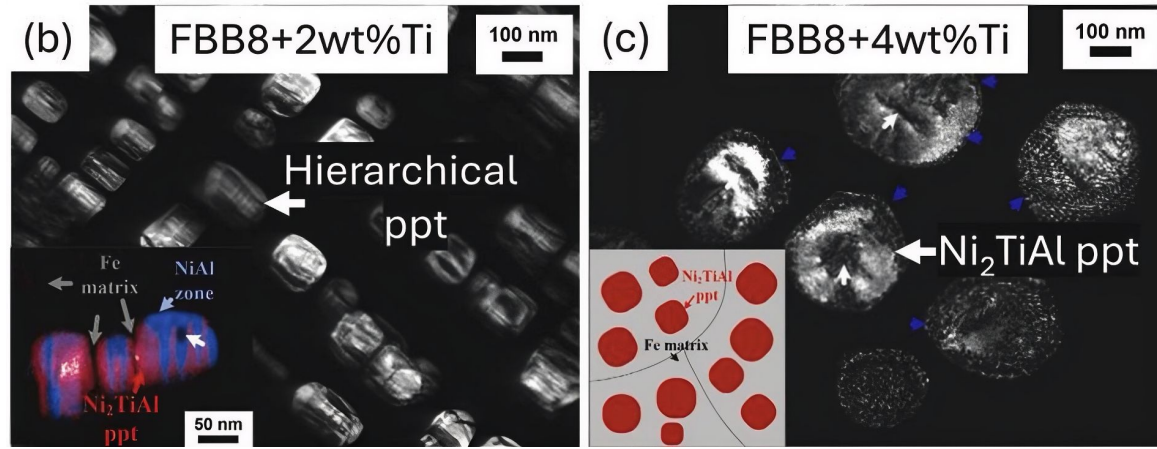


TEM image of FBB8 alloy with Ti=0. Treated condition: 1200°C for 0.5 h and aged at 700 °C for 100 h, followed by air cooling.

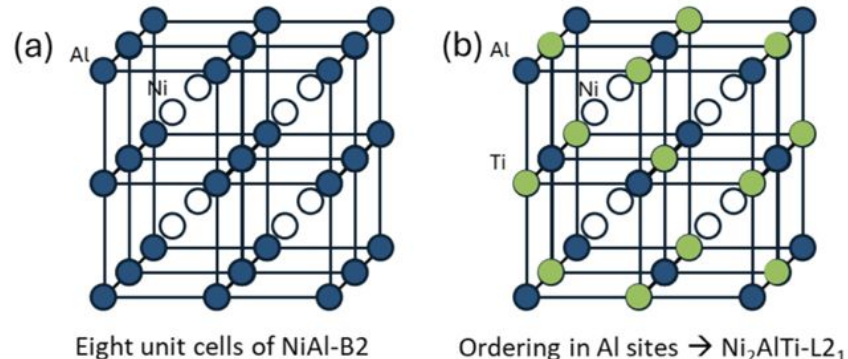
Fe-based BCC Lattice Misfit

Hierarchical Microstructure (Moderate To High Lattice Misfit)

- Ti raises misfit to 0.67-1.39% (moderate coherency strain)
- L2_1 domains (5-20 nm) inside B2 precipitates (50-200 nm) → create dual-barrier energy landscape → Strong elastic interaction and microstructural stability.
- Excessive Ti over 4 wt%:
 - Complete B2 → L2_1 transformation
 - Loss of coherency → reduce creep threshold stress



TEM image of FBB8 alloy with b) Ti=2%; c) Ti=4%. Treated condition: 1200°C for 0.5 h and aged at 700 °C for 100 h, followed by air cooling.

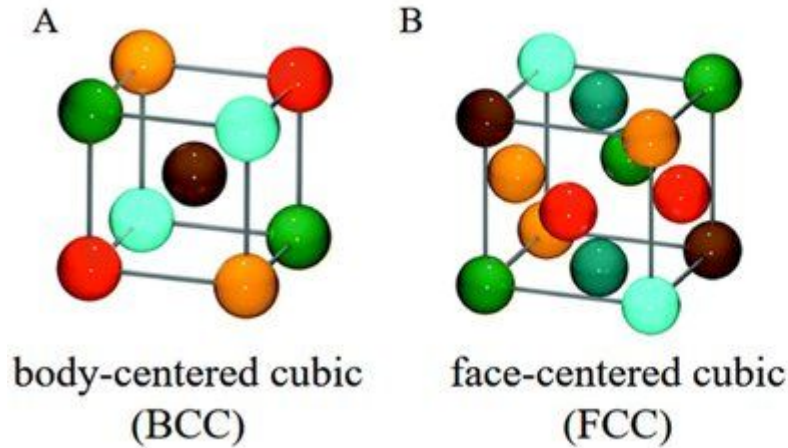


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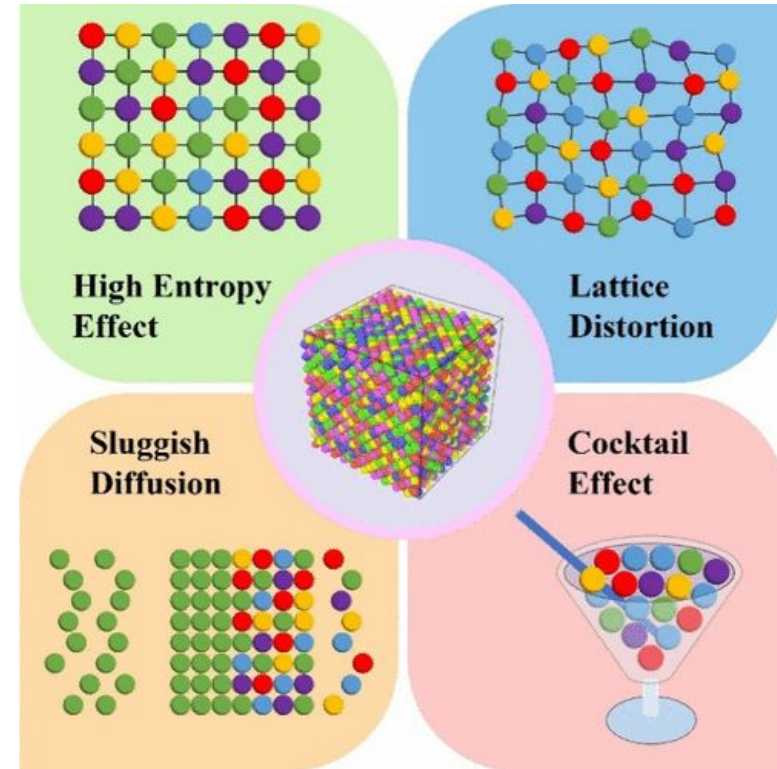
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RHEA Superalloys Core Effects

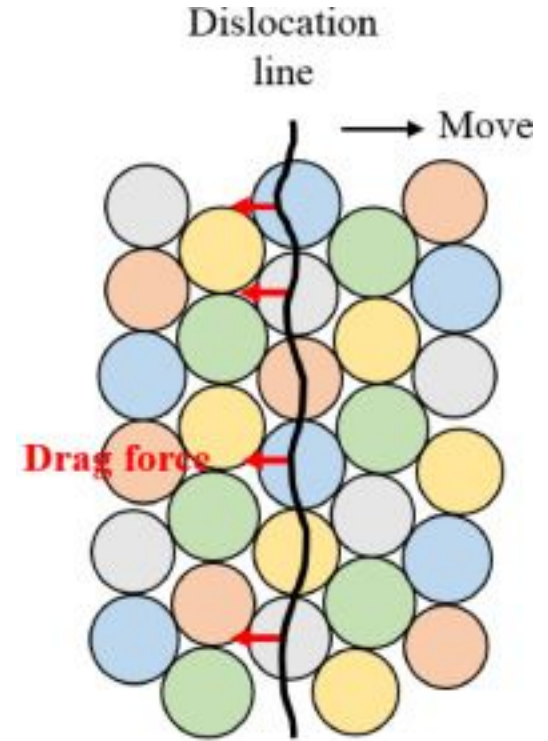
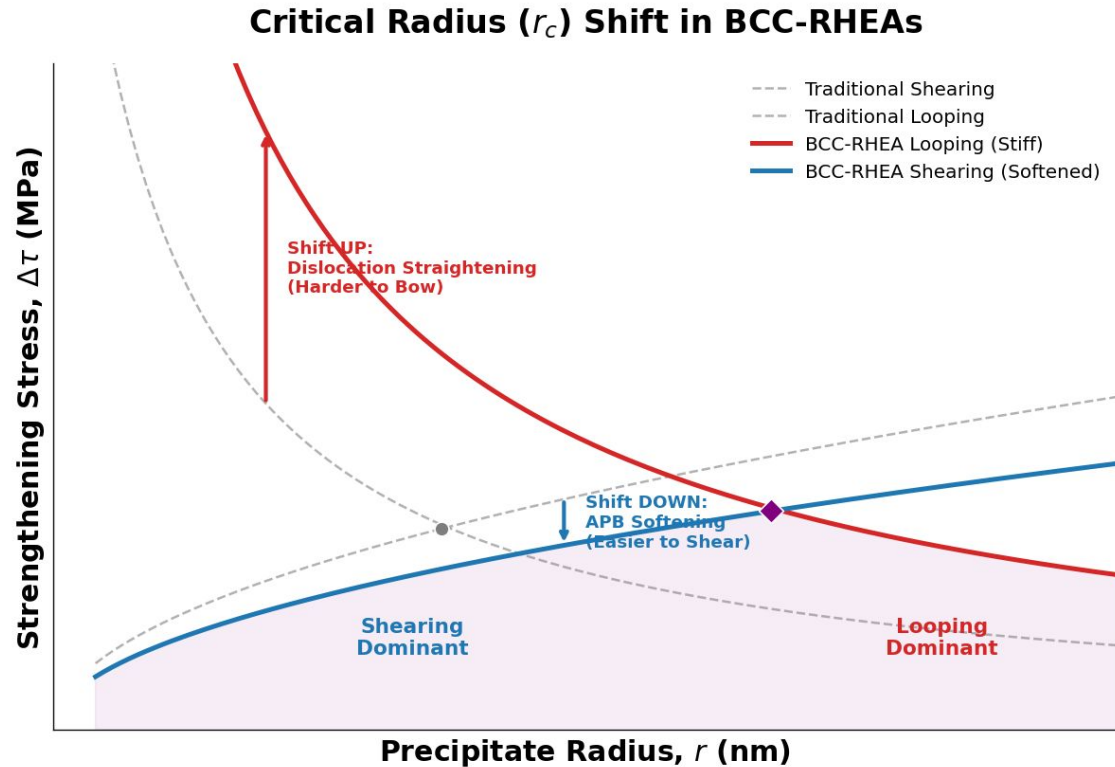


- RHEA: 4 or more elements
- Break **Hume-Rothery rules** for alloy designing
- Sluggish diffusion is not true in relative scale (compared to T_m)

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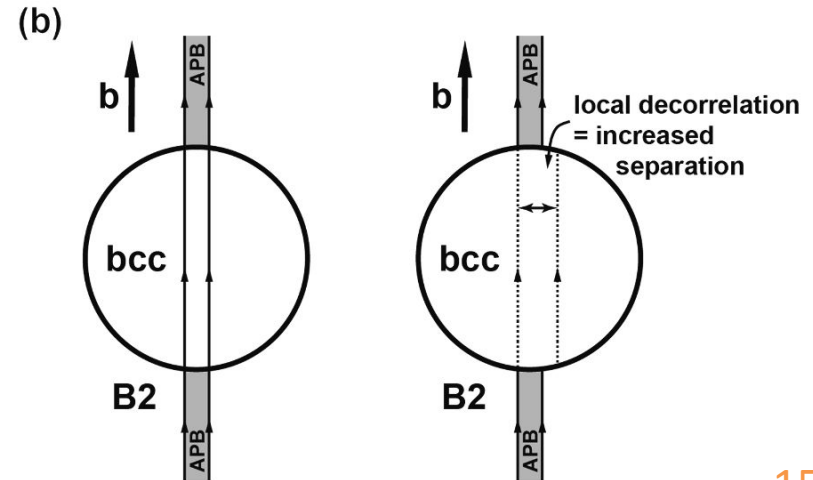
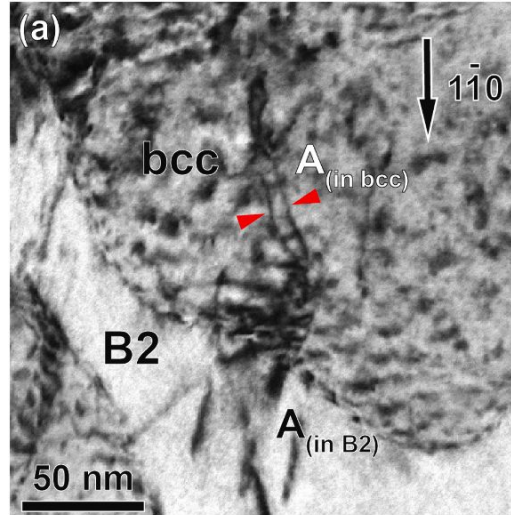
Matrix Ruggedness (1): Dislocation movement



Matrix Ruggedness (2): Entropy-Stabilized Ductility

- In traditional alloys, APB energy is high. RHEAs reduce APB energy.
- Rugged matrix acts as an obstacle for dislocations in the B2 matrix while encouraging dislocation mobility in precipitates.
- Ductility originates from dislocation movement, therefore, is increased.

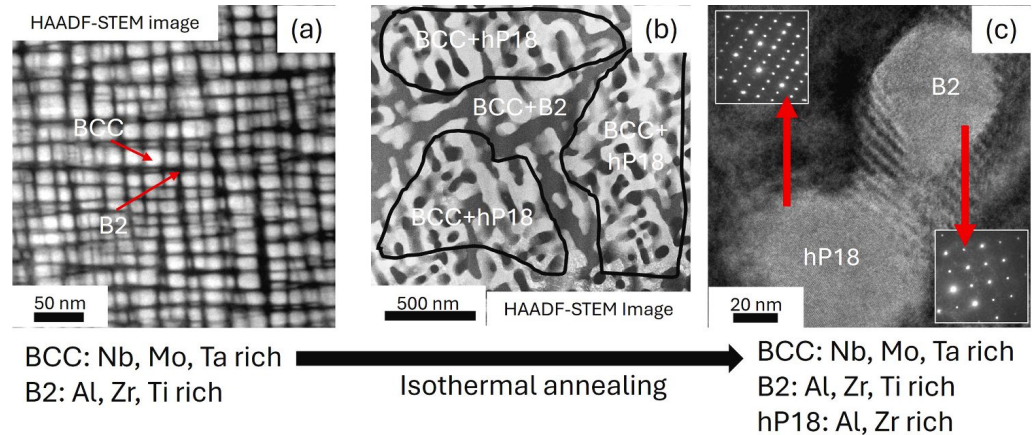
AlMo_{0.5}NbTa_{0.5}TiZr
paired dislocations
were observed cutting
through 60 nm
coherent precipitates.



Pathway Engineering

- Invert topology, where continuous matrix is now the ordered brittle phase, implies ductility drops significantly
- To prevent the phenomena:
 - Controlling ordered phase forming elements like Al, Ti, etc.
 - Utilize nucleation & growth instead of spinodal decomposition
 - Heat treatment

Heat treating BCC+B2 superalloy microstructure into three-phase BCC + B2 + ω' (AlZr-rich hP18 phase)



Engineering Challenges & Outlooks

- Pesting - Volatile/Liquidize phases: Refractory phases can be volatile/liquidized easily at high temperature => Adding **sufficient Al**
- 3D manufacturing leads to formation 'hot tearing' cracks
=> Substituting Zr with Hf can stabilize desired phase with narrower freezing range, allow defect free 3D printing.
- Computational tools and Machine Learning can be employed to explore the massive space in RHEAs to tune for the optimized alloys

Thank You